LESSON 9.3

Write Equations to Represent Relationships Between Two Variables

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation from a table or a graph to model the relationship between the dependent and independent variables.

WHY THIS MATTERS Miguel is interested in joining a recreation center. Recreation World charges the monthly fee shown on the flyer. How can Miguel determine the cost of being a member of Recreation World for any number of months?

REMEMBER

Write Equations to Represent Relationships Between Two Variables

Writing Two-Variable Equations. Formula: y = kx | Dependent = (Unit Rate) × Independent A table can only list some values. An equation captures the whole pattern at once, so it works for any number of months or hours.

1. Identify the independent variable (x) and dependent variable (y).
2. Find the constant unit rate (k = y ÷ x).
3. Write the equation relating the two variables: y = kx (or y = x + c)

Words: equation *(ecuación)* · variable *(variable)* · rate *(tasa)* · represent *(representar)* · define the variables *(definir las variables)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each equation by what it really says.

Groups: Says 'y is 4 times x' · Says something different

_____ y = 4x

_____ y = x × 4

_____ x = y ÷ 4

_____ y = x + 4

_____ x = 4y

_____ y = 4 ÷ x

HINT Test with a value: if x = 2 and y is 4 times x, then y must be 8. Which equations agree?

2 **Complete the table. Show how you found each value.**

A canoe rents for $15 per hour: c = 15h. Complete the table.

Hours (h)	Substitute	Cost in dollars (c)
2		
5		
8		

HINT Replace h with the number of hours, then multiply by the unit rate 15.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

In the ordered pair (x, y), which value is graphed along the horizontal axis?

(A) y

(B) Both of them

(C) Neither one

(D) x

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A table pairs x = 1, 2, 3 with y = 3, 6, 9. Which ordered pair is in the table?

(A) (2, 6)

(B) (2, 3)

(C) (3, 2)

(D) (6, 2)

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

On a graph of hours worked and money earned, which axis shows the independent variable?

(A) The vertical axis

(B) The horizontal axis

(C) Either axis works

(D) Neither axis

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Both equations charge by a rate. What do m and h each stand for in their own situation, and why would c = 24.95m be wrong for the canoe?

SENTENCE STARTER In ___, ___ stands for ___, so the rate ___ means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 9.3

Write Equations to Represent Relationships Between Two Variables

Set B · Version B · Second form · core practice

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LEARNING TARGET I can write an equation from a table or a graph to model the relationship between the dependent and independent variables.

WHY THIS MATTERS Miguel is interested in joining a recreation center. Recreation World charges the monthly fee shown on the flyer. How can Miguel determine the cost of being a member of Recreation World for any number of months?

1 **Circle the letter of the best answer. Show how you know.**

A table shows x = 0, 1, 2, 3 and y = 5, 7, 9, 11. How much does y change each time x increases by 1?

(A) It increases by 1
(B) It increases by 2
(C) It increases by 5
(D) It increases by 7

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Diamond is considering using an online music service. She can either purchase songs one at a time or pay to stream songs. Each choice has its own relationship between the number of songs and the total cost. In Diamond's situation, what should be the independent variable in both equations?

(A) the total cost
(B) the price per song
(C) the type of service (purchase or stream)
(D) the number of songs

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Diamond is considering using an online music service. She can either purchase songs one at a time or pay to stream songs. Each choice has its own relationship between the number of songs and the total cost. Why does Diamond need TWO separate equations instead of one equation for both options?

(A) Purchasing and streaming likely follow different cost rules (such as a per-song price vs. a flat rate), so each relationship between songs and cost is different

(B) The number of songs is different for each option

(C) One equation can only have one variable

(D) Streaming has no relationship between songs and cost

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Diamond is considering using an online music service. She can either purchase songs one at a time or pay to stream songs. Each choice has its own relationship between the number of songs and the total cost. Once Diamond has both equations, how does she use them to decide which option is better?

(A) Add the two equations together to find one combined cost

(B) Pick whichever equation has the smaller numbers written in it, without solving anything

(C) Substitute the same number of songs into both equations and compare the two total costs

(D) Graph only one equation and assume the other follows the same line

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Recreation World charges $24.95 per month. Let m = number of months and c = total cost. Which equation represents the relationship?

(A) $c = 24.95m$

(B) $c = 24.95 + m$

(C) $m = 24.95c$

(D) $c = m \div 24.95$

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Both equations charge by a rate. What do m and h each stand for in their own situation, and why would c = 24.95m be wrong for the canoe?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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